Unit 3

Lesson Three

The definite integral

The fundamental theorem of calculus

If the function f is continuous on the interval $[a , b]$, and F is any anti derivative to the function f on the same interval , then :

$$\int_a^b f(x)\, dx = F(b) - F(a)$$

- $\int_a^b f(x)\, dx$ is called the definite integral and read as "the definite integral of $f(x)$ with respect to x from a to b" and a , b are called the lower and upper limits of integration.
- The symbol of the variable x can be replaced by another symbol without affecting the amount of the integration.

  i.e. : $\int_a^b f(x)\, dx = \int_a^b f(y)\, dy = \int_a^b f(z)\, dz$ ....

- $F(b) - F(a)$ is expressed in the form $[F(x)]_a^b$ or $F(x)\Big|_a^b$
- The definite integral is a real number , whose value depends on

  1. The upper and lower terms of the definite integration

  i.e. On the two numbers a and b respectively.

  2. The rule of the function f

- You can get the definite integral by finding the indefinite integral with disregarding the constant of the integration then , you can substitute the variable by the two terms of the integration.

Notice that

Disregarding the constant because

$$\int_a^b f(x)\, dx = [F(x) + c]_a^b = [F(b) + c] - [F(a) + c] = F(b) - F(a) + c - c = F(b) - F(a)$$

- All the rules of the indefinite integrations and the table of the standard integrations are to be applied to find the value of the definite integration of a continuous function.

If f , g are two continuous functions on the interval $[a , b]$

1. $\int_a^b [f(x) \pm g(x)]\, dx = \int_a^b f(x)\, dx \pm \int_a^b g(x)\, dx$
2. $\int_a^b k f(x)\, dx = k \int_a^b f(x)\, dx$ (where $k \in \mathbb{R}$)

Theorem

If the function f is continuous on the interval $[a , b]$, then it is integrable on this interval.

Notice that

The previous theorem shows that the continue is a sufficient condition to integrate but it is not necessary , which means that every continuous function on an interval is integrable on this interval but not every integrable function on an interval is a continuous function on this interval.

The difference between the definite integral and indefinite integral

1. $\int f(x)\, dx$ "an indefinite integral" is a family of functions , the derivative of each gives $f(x)$
2. $\int_a^b f(x)\, dx$ "definite integral" is a real number.

Properties of definite integral

If f is a continuous on $[a , b]$, $c \in]a , b[$, then :

1. $\int_b^a f(x)\, dx = -\int_a^b f(x)\, dx$
2. $\int_a^a f(x)\, dx = \text{zero}$
3. $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$

Example 1

Find the definite integral of the function f ; from $x = -1$ to $x = 3$, where $f(x) = 4x^3 + 1$

Solution

$\because$ The function f is a polynomial function.

$\therefore$ The function is continuous on $\mathbb{R}$

$\therefore \int_{-1}^{3} (4x^3 + 1)\, dx = [x^4 + x]_{-1}^{3} = [(3)^4 + (3)] - [(-1)^4 + (-1)] = 84$

* It is possible to use the scientific calculator to find the result as follow :



Example 2

Find the value of each of the following :

1. $\int_1^2 (3x^2 + 2x - 1)\,dx$

2. $\int_2^5 \frac{2}{\sqrt{x-1}}\,dx$

3. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\,dx$

4. $\int_0^{\frac{\pi}{4}} \sec\theta \tan\theta\,d\theta$

5. $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{dx}{\cos^2 x}$

Solution

1. $\int_1^2 (3x^2 + 2x - 1)\,dx = [x^3 + x^2 - x]_1^2$

$= [(2)^3 + (2)^2 - (2)] - [(1)^3 + (1)^2 - (1)] = 9$

2. $\int_2^5 \frac{2}{\sqrt{x-1}}\,dx = \int_2^5 2(x-1)^{-\frac{1}{2}}\,dx$

$= [4(x-1)^{\frac{1}{2}}]_2^5$

$= [4(4)^{\frac{1}{2}}] - [4(1)^{\frac{1}{2}}] = 4$

3. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\,dx = [\sin x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= [\sin\frac{\pi}{2}] - [\sin\frac{-\pi}{2}]$

$= 1 - (-1) = 2$

4. $\int_0^{\frac{\pi}{4}} \sec\theta \tan\theta\,d\theta = [\sec\theta]_0^{\frac{\pi}{4}}$

$= [\sec\frac{\pi}{4}] - [\sec 0]$

$= \sqrt{2} - 1$

5. $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{dx}{\cos^2 x} = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec^2 x\,dx = [\tan x]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$

$= [\tan\frac{\pi}{3}] - [\tan\frac{\pi}{4}]$

$= \sqrt{3} - 1$

Notice as using calculator that

In case of performing an integration operation with a variable represents an angle we know that variable is in radian , then we have to use in calculators one of the following 2 methods.

Method 1

We have to change the mode of calculator into the Rad mode

Method 2

We use the common mode Deg with following the angle by pressing SHIFT Ans 2

For example to calculate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx$ we press the following bottons

Start

SHIFT Mode 4 $\int$ Cos

ALPHA ×)) REPLAY (−) SHIFT

π ×10^x 2 REPLAY REPLAY SHIFT

π ×10^x 2 =

Or

Start

$\int$ Cos ALPHA ×) SHIFT

Ans 2) REPLAY (−) SHIFT

π ×10^x 2 REPLAY REPLAY

SHIFT π ×10^x 2 =

The indicated answer will be "2"

Example 3

Find the value of each of the following :

1. $\int_1^e \frac{dy}{y}$ | **2.** $\int_{-3}^{-1} \frac{2}{x} dx$ | **3.** $\int_0^{\ln 5} e^x dx$

Solution

1. $\int_1^e \frac{dy}{y} = [\ln|y|]_1^e = [\ln e] - [\ln 1] = 1 - 0 = 1$

2. $\int_{-3}^{-1} \frac{2}{x} dx = 2[\ln|x|]_{-3}^{-1} = 2[\ln|-1| - \ln|-3|] = -2 \ln 3$

3. $\int_0^{\ln 5} e^x dx = [e^x]_0^{\ln 5} = e^{\ln 5} - e^0 = 5 - 1 = 4$

Remember

$e^{\ln a} = a$

Example 4

If $\int_1^5 f(x)\,dx = 8$, $\int_5^1 g(x)\,dx = 3$

Find the value of $\int_1^5 [f(x) + 3g(x) - 2]\,dx$

Solution

$\because \int_5^1 g(x)\,dx = 3$

$\therefore \int_1^5 g(x)\,dx = -3$

$\therefore \int_1^5 [f(x) + 3g(x) - 2]\,dx = \int_1^5 f(x)\,dx + 3\int_1^5 g(x)\,dx - \int_1^5 2\,dx$

$= 8 + 3 \times (-3) - [2x]_1^5 = 8 - 9 - [10 - 2] = -9$

Example 5

If f is a continuous function on $\mathbb{R}$, $\int_1^4 f(x)\,dx = 7$, $\int_6^4 f(x) = -12$,

Find $\int_1^6 f(x)\,dx$

Solution

$\because f$ is continuous function on $\mathbb{R}$, $x = 4$ divides the interval $[1\ ,6]$

$\therefore \int_1^6 f(x)\,dx = \int_1^4 f(x)\,dx + \int_4^6 f(x)\,dx$

$= \int_1^4 f(x)\,dx - \int_6^4 f(x)\,dx = 7 - (-12) = 19$

Example 6

Find : $\int_{-2}^{2} f(x)\,dx$ where $f(x) = \begin{cases} 2x^3 + 3 & , \ x \le -1 \\ 3x + 4 & , \ x > -1 \end{cases}$

Solution

Because $f(-1^+) = f(-1^-) = f(-1) = 1$

$\therefore f$ is integrable on the interval $[-2\ ,2]$

Notice that

f is continuous on $[-2\ ,2]$

$$\therefore \int_{-2}^{2} f(x)\,dx = \int_{-2}^{-1} (2x^3+3)\,dx + \int_{-1}^{2} (3x+4)\,dx$$

$$= \left[\frac{1}{2}x^4 + 3x\right]_{-2}^{-1} + \left[\frac{3}{2}x^2 + 4x\right]_{-1}^{2}$$

$$= \left(-2\frac{1}{2} - 2\right) + \left(14 + 2\frac{1}{2}\right) = 12$$

Example 7

Find the value of each of the following :

1. $\int_{0}^{6} |x-4|\,dx$ **2.** $\int_{-2}^{4} |x^2-3x|\,dx$

Solution

1. $\because |x-4| = \begin{cases} x-4 & , \ x \geq 4 \\ -x+4 & , \ x < 4 \end{cases}$

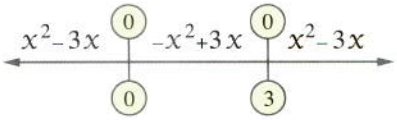
, $\because f$ is continuous at $x = 4$

$$\therefore \int_{0}^{6} |x-4|\,dx = \int_{0}^{4} (-x+4)\,dx + \int_{4}^{6} (x-4)\,dx$$

$$= \left[-\frac{1}{2}x^2 + 4x\right]_{0}^{4} + \left[\frac{1}{2}x^2 - 4x\right]_{4}^{6} = [8-0] + [-6-(-8)] = 10$$

2. Put $f(x) = x^2 - 3x = 0$

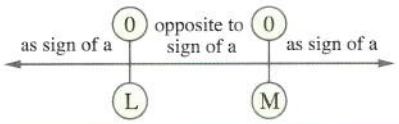
$\therefore x(x-3) = 0$ $\therefore x = 0$, $x = 3$



$$\therefore |x^2-3x| = \begin{cases} x^2-3x & , \ x \leq 0 \\ -x^2+3x & , \ 0 < x \leq 3 \\ x^2-3x & , \ x > 3 \end{cases}$$

Remember that

If the function $f : f(x) = a x^2 + b x + c$ has two real different roots L , M where L < M , then : $f(x)$ has the same sign as a (coefficient of x^2) when $x \in \mathbb{R} - [L, M]$ and it has a sign opposite to a when $x \in]L, M[$ and $f(x) = 0$ when $x \in \{L, M\}$



$$\therefore \int_{-2}^{4} |x^2-3x|\,dx = \int_{-2}^{0} (x^2-3x)\,dx + \int_{0}^{3} (-x^2+3x)\,dx + \int_{3}^{4} (x^2-3x)\,dx$$

$$= \left[\frac{1}{3}x^3 - \frac{3}{2}x^2\right]_{-2}^{0} + \left[-\frac{1}{3}x^3 + \frac{3}{2}x^2\right]_{0}^{3} + \left[\frac{1}{3}x^3 - \frac{3}{2}x^2\right]_{3}^{4}$$

$$= \left[0 + 8\frac{2}{3}\right] + \left[4\frac{1}{2} - 0\right] + \left[-2\frac{2}{3} + 4\frac{1}{2}\right] = 15$$

Properties of definite integral of odd functions and even functions

1. If the function f is continuous and odd on the interval $[-a, a]$, then :

$$\int_{-a}^{a} f(x)\,dx = \text{zero}$$

2. If the function f is continuous and even on the interval $[-a, a]$ then :

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

Example 8

Evaluate each of the following :

1. $\int_{-3}^{3} \frac{x^5 + \sin x}{x^2 + 1}\,dx$ **2.** $\int_{-2}^{2} (x^4 - 1)\,dx$

Solution

1. $\because f(x) = \frac{x^5 + \sin x}{x^2 + 1}$ is continuous on $\mathbb{R}$

, $\because f(-x) = \frac{(-x)^5 + \sin(-x)}{(-x)^2 + 1} = \frac{-x^5 - \sin x}{x^2 + 1} = -\left(\frac{x^5 + \sin x}{x^2 + 1}\right) = -f(x)$

$\therefore$ The function is odd

$\therefore \int_{-3}^{3} \frac{x^5 + \sin x}{x^2 + 1}\,dx = \text{zero}$

2. $\because f(x) = x^4 - 1$ is a polynomial function

$\therefore f$ is continuous on $\mathbb{R}$

, $\because f(-x) = (-x)^4 - 1 = x^4 - 1 = f(x)$

$\therefore$ The function f is even.

Remember

If each of (x) and $(-x)$ belonges to domain of the function f , then

* In the even function $f(-x) = f(x)$
* In odd function $f(-x) = -f(x)$

$$\therefore \int_{-2}^{2} (x^4 - 1)\,dx = 2\int_{0}^{2} (x^4 - 1)\,dx = 2\left[\frac{x^5}{5} - x\right]_0^2$$

$$= 2\left[\frac{22}{5} - 0\right]_0^2 = \frac{44}{5} = 8.8$$

Example 9

If the function f is an even function , continuous on the interval $[-5 , 5]$, $\int_{-5}^{5} f(x)\,dx = 30$, $\int_{0}^{3} f(x)\,dx = 9$, find the value of : $\int_{-5}^{-3} f(x)\,dx$

Solution

$\because$ The function f is even on the interval $[-5 , 5]$

, $\because \int_{-5}^{5} f(x)\,dx = 30$

$\therefore 2\int_{0}^{5} f(x)\,dx = 30$

$\therefore \int_{0}^{5} f(x)\,dx = 15$

$\therefore \int_{0}^{3} f(x)\,dx + \int_{3}^{5} f(x)\,dx = 15$

$\therefore 9 + \int_{3}^{5} f(x)\,dx = 15$

$\therefore \int_{3}^{5} f(x)\,dx = 6$

$\therefore \int_{-5}^{-3} f(x)\,dx = 6$

Notice that

1. If f is an even function and a , b are two positive numbers , then $\int_{a}^{b} f(x)\,dx = \int_{-b}^{-a} f(x)\,dx$
2. If f is an odd numbers and a , b are two positive numbers , then $\int_{a}^{b} f(x)\,dx = -\int_{-b}^{-a} f(x)\,dx$

Example 10

Evaluate each of the following :

1. $\int_{-3}^{-2} x(x+2)^3\,dx$
2. $\int_{0}^{\sqrt{5}} x\sqrt{x^2+4}\,dx$

Solution

Notice that

The solution using the substitution method.

1. Put $z = x + 2$

$\therefore x = z - 2$, $dx = dz$

$\therefore \int x(x+2)^3\,dx = \int (z-2) \times z^3\,dz = \int (z^4 - 2z^3)\,dz$

$= \frac{1}{5}z^5 - \frac{1}{2}z^4 + c = \frac{1}{10}z^4(2z-5) + c$

$= \frac{1}{10}(x+2)^4\left(2(x+2)-5\right) + c = \frac{1}{10}(x+2)^4(2x-1) + c$

$\therefore \int_{-3}^{-2} x(x+2)^3\,dx = \left[\frac{1}{10}(x+2)^4(2x-1)\right]_{-3}^{-2} = \text{zero} - \left[\frac{1}{10} \times 1 \times -7\right] = \frac{7}{10}$

* It is possible to find the solution directly by using the corresponding values of z to the lower and upper terms of the original integration $[X = -3 \ , \ X = -2]$

at $X = -3$, then $z = -1$, at $X = -2$, then $z = 0$

$$\therefore \int_{-3}^{-2} X (X+2)^3 \, dX = \int_{-1}^{0} (z^4 - 2z^3) \, dz$$

$$= \left[\frac{1}{5} z^5 - \frac{1}{2} z^4\right]_{-1}^{0} = \text{zero} - \left[-\frac{1}{5} - \frac{1}{2}\right] = \frac{7}{10}$$

2. Put $z = X^2 + 4$ $\quad \therefore dz = 2X \, dX$

$$\therefore \int X \sqrt{X^2+4} \, dX = \frac{1}{2} \int \sqrt{X^2+4} \times 2X \, dX = \frac{1}{2} \int \sqrt{z} \, dz$$

$$= \frac{1}{2} \int z^{\frac{1}{2}} dz = \frac{1}{2} \times \frac{2}{3} z^{\frac{3}{2}} + c = \frac{1}{3} \sqrt{(X^2+4)^3} + c$$

$$\therefore \int_{0}^{\sqrt{5}} X \sqrt{X^2+4} \, dX = \left[\frac{1}{3} \sqrt{(X^2+4)^3}\right]_{0}^{\sqrt{5}} = \frac{27}{3} - \frac{8}{3} = \frac{19}{3}$$

* It is possible to find the solution directly by using the corresponding values of z to the lower and upper terms of the original integration

at $X = 0$, then $z = 4$, at $X = \sqrt{5}$, then $z = 9$

$$\therefore \int_{0}^{\sqrt{5}} X \sqrt{X^2+4} \, dX = \frac{1}{2} \int_{4}^{9} z^{\frac{1}{2}} \, dz = \left[\frac{1}{3} z^{\frac{3}{2}}\right]_{4}^{9} = \frac{27}{3} - \frac{8}{3} = \frac{19}{3}$$

Important Remarks

1. In the indefinite integral

$\int f(X) \, dX - \int f(X) \, dX = \text{constant}.$

But in the definite integral.

$$\int_{a}^{b} f(X) \, dX - \int_{a}^{b} f(X) \, dX = 0$$

2. Notice the difference between these two answers :

* $\int_{1}^{2} \frac{1}{X^2} dX = \left[\frac{-1}{X}\right]_{1}^{2} = -\frac{1}{2} - (-1) = \frac{1}{2}$ "Correct answer"

* $\int_{-2}^{2} \frac{1}{X^2} dX = \left[\frac{-1}{X}\right]_{-2}^{2} = -\frac{1}{2} - \frac{1}{2} = -1$ "Incorrect answer"

thats because evaluating definite integral of a discontinuous function in the interval of integration is beyond the scope of our study , so « $\int_{-2}^{2} \frac{1}{X^2} dX$ can not be found».